

Lab 01 - Learning as Compression & MDL

Fundamentos de Aprendizagem Automática

Mário Antunes

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Laboratory Guide 01: Information, Compression, and Learning

1. Objectives

- Understand the mathematical connection between learning, information, and data compression.
- Compute empirical Shannon Entropy on discrete distributions.
- Implement polynomial curve fitting and analyze the trade-off between model complexity and residual fit.
- Apply the Minimum Description Length (MDL) principle / Bayesian Information Criterion (BIC) to select optimal models without a validation set.

2. Theoretical Background

2.1 Shannon Entropy

Given a discrete random variable X with alphabet $\mathcal{X}$ and probability mass function $p(x)$:

$$H(X) = - \sum_{x \in \mathcal{X}} p(x) \log_2 p(x)$$

Entropy defines the fundamental limit of lossless compression.

2.2 Minimum Description Length (Two-Part Code)

For a model with k parameters evaluated on n samples with mean squared error MSE:

$$\text{MDL} \approx \frac{k}{2} \log_2(n) + \frac{n}{2} \log_2(\text{MSE})$$

Notice that as k increases, the first term (model cost) grows logarithmically with n , while the second term (data cost given the model) decreases.

3. Practical Exercises

Exercise 1: Computing Entropy of Text and Noise

1. Open notebook `notebooks/01-compression/01_entropy.ipynb`.
2. Generate three sequences:
 - Structured repetitive text (e.g. ABABABAB . . .).
 - Natural English text sample.
 - Uniformly distributed random bytes.
3. Compute the empirical entropy of each sequence and compare against the file size after gzip/zlib compression.

Exercise 2: Polynomial Regression and Overfitting

1. Generate synthetic noisy data:

$$y = \sin(2\pi x) + \epsilon, \quad \epsilon \sim \mathcal{N}(0, 0.15^2)$$

with $n = 30$ samples in $[0, 1]$.

2. Fit polynomials of degree $d \in [1, 15]$.
3. Observe how higher-degree polynomials achieve zero training MSE but wildly oscillate between data points.

Exercise 3: Automated Model Selection via MDL / BIC

1. Calculate the MDL score for each polynomial degree $d \in [1, 15]$.
2. Identify the minimum of the MDL curve.
3. Verify that the model selected by MDL corresponds to the degree that maximizes test generalization.